Jawad Mohammed

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Professional Summary

Software engineer with 6+ years building fault-tolerant, high-availability distributed systems in Java, Spring Boot, Kafka and Kubernetes. Migrated a Verizon IoT platform from COBOL to Java microservices, and moved 10+ Jenkins CI pipelines to AWS. Strengthens production reliability through root cause analysis, monitoring controls and CI/CD quality gates, with 15 consecutive months of zero critical incidents. Recent personal projects add GitOps on Kubernetes with Terraform and Argo CD, chaos-tested event pipelines and observability with Prometheus and Grafana. MS in Computer Science at the University of North Texas, graduating December 2026.

Technical Skills

Languages: Java, Python, SQL, JavaScript

Platform and DevOps: Kubernetes, Docker, Terraform, Argo CD, GitOps, Gateway API, Jenkins, GitHub Actions, CI/CD, Maven, Gradle, Git

Cloud: AWS (S3, EC2, DynamoDB, Lambda, Step Functions); EKS and IRSA defined in Terraform

Reliability and observability: Prometheus, Grafana, load testing, chaos testing, autoscaling, root cause analysis, incident prevention

Backend and data: Spring Boot, Spring Batch, Spring Cloud, Spring Security, JPA, Hibernate, Apache Kafka, Kafka Streams, PostgreSQL, Redis, MongoDB, DynamoDB

Architecture: Distributed systems, microservices, event-driven architecture, REST APIs, high availability, system design, data structures and algorithms

Professional Experience

Senior Associate Consultant | Infosys (Client: Verizon IoT), India Apr 2023 - Nov 2024

- Contributed to the design and rollout of 4 Spring Boot microservices for IoT event ingestion during a platform migration off legacy COBOL onto Java, increasing system throughput by 20% while handling approximately 10K events per cycle.
- Owned the Kafka Streams integration, building event-driven pipelines that sustained 10K+ events per minute at peak device load with reliable, low-latency delivery.
- Refactored legacy COBOL business logic into Java service layers using object-oriented design, removing procedural bottlenecks and improving maintainability and reliability.
- Built Spring Batch pipelines for large-scale dataset processing with fault-tolerant exception handling, ensuring 100% data integrity by preventing single bad records from stalling entire jobs.
- Took part in code reviews and system design discussions, contributing to team engineering standards.

Tech: Java, Spring Boot, Apache Kafka, Kafka Streams, PostgreSQL, Redis, Kubernetes, Spring Batch, Docker

Tech Intern, Associate Developer QA, Developer QA | Thomson Reuters, India Sep 2018 - Mar 2023

- Led the migration of 10+ Jenkins CI pipelines to AWS after recurring job failures blocked releases, keeping release schedules uninterrupted and reducing release cycle time by 30%.
- Drove root cause analysis on critical production incidents and added preventive monitoring and validation controls, reducing repeat outages by 15%.
- Maintained the production environment with zero critical incidents across 15 consecutive months during a staffing shortage.
- Architected an API validation framework with CI/CD enforcement during the Spring Boot microservices migration, catching integration failures before production and reducing defect rates by 30%.
- Found PostgreSQL performance degradation at 10K+ transaction volumes, then led a proof of concept with AWS S3 and DynamoDB that validated 25% throughput gains under realistic load.
- Automated 40% of manual regression testing with JUnit, Mockito, Mocha and Cucumber in CI/CD pipelines, while maintaining 100% global statutory compliance accuracy.
- Promoted from intern to Associate Developer within 10 months on the Financial Reporting platform.

Tech: Java, Spring Boot, PostgreSQL, Redis, JPA/Hibernate, REST APIs, AWS (S3, EC2, DynamoDB), Jenkins, Docker, JUnit, Mockito, Cucumber

Graduate Teaching Assistant | University of North Texas, Denton, TX Aug 2025 - Present

- Support a 100-student Foundations of Computing course through weekly sessions, 6 hours of weekly office hours, grading and exam proctoring.

Projects

EKS GitOps Platform | *Terraform, Kubernetes, Argo CD, Gateway API, autoscaling*

- Defined VPC, EKS and IRSA in Terraform, validated by 12 offline tests against a mocked AWS provider; Argo CD deploys every change from Git.
- On a local Kubernetes cluster: 14.6 s median from git push to all pods updated, and 1.4 s rollback with git revert.
- Kept 99.997% request success during releases under load; a 3x traffic spike scaled pods from 2 to 6 in 69 s.
- Cut fast 503 errors during spikes from 21% to 2.7% through three fixes; designed scoped CI roles, break-glass access and per-tenant isolation.

IoT Telemetry Pipeline | *Kafka, stream processing, PostgreSQL, Redis*

- Rebuilt a polling design as an event pipeline with real-time device status and offline detection from missing heartbeats.
- Sustained 500 events/sec at about 50 ms p95, and 2,000 events/sec with no loss.
- Lost zero acknowledged events across 2.5 million while killing a broker, consumer, stream processor and ingest node under load.
- Benchmarked batch inserts 29% faster than plain JDBC batching and 87% faster than a row-by-row loop.

PulseLink Real-Time Alerting | *Java, WebSockets, Prometheus, Grafana, Docker*

- Built streaming patient-alert delivery with escalation, an audit trail and authenticated WebSocket dashboards; 47 tests pass.
- Found and fixed a latency bug through load testing, cutting p99 from 1.5 s to 0.13 s.
- Raised throughput 24% (4,334 to 5,377 events/sec) and cut p95 alert delay during a backlog from 8.2 s to 0.5 s.
- At 3x traffic, held 154 ms p95 with no alerts lost, monitored with Prometheus and Grafana.

Publications

Mohammed, J., and Hossain, G. "Detecting Race Conditions in FHIR-Based Healthcare Interoperability: A Formal Graph-Theoretic Approach." Accepted for publication, MDPI, 2026.

- Modelled concurrent FHIR API access as a directed graph (FRAG) and formally defined three race-condition classes with soundness and completeness proofs.
- Built a Python/NetworkX detector reaching 90.0% F1 against 25.5% for a time-window baseline on 1,500 synthetic FHIR R4 logs, scaling near-linearly to 250,000 events.

Education

Master of Science in Computer Science Engineering | University of North Texas, Denton, TX Dec 2026

Master of Technology in Computer Science Engineering | Vasavi College of Engineering, India Aug 2019

Bachelor of Engineering in Information Technology | Deccan College of Engineering and Technology, India Jul 2017